

Capacity Win: Freeing Bed Days by Trimming Long Stays

This Project turns a raw admissions dataset into a one-page, decision-ready dashboard that shows where a small cut to long stays frees the most capacity.

I:

- Ranked Medical conditions by bed days to pick a high-impact focus.
- Within the chosen condition, compared the median and 90% percent (p90) length of stay (LOS) by admission type to find the long stay tail.
- Sized the prize with a simple KPI: bed days freed per week from a modest p90 reduction (10%/15%/20%).
- Added context: Age Groups that consume the most bed days and weekday spikes when new bed days start (so operations can staff to demand).

Headline: Focusing on the Emergency pathway for the target condition yields the largest weekly capacity win because of higher admission volume, even when p90 is similar elsewhere.

Business Problem

Hospitals struggle with congested beds, slow flow, and elective backlogs. We are not adding beds: we are freeing capacity by trimming the long stay tail. The guiding question is:

“Where does a small reduction in 90 percent Length of stay (p90 LOS) free the most bed days each week?”

We answer this with five practical questions that fit on a single Power BI page and lead directly to a two-week pilot.

Business Questions

N/O	Question	Department Head	Objective	Recommended Chart
1	Which medical condition accounts for the most total bed days in the selected period?	COO / Operations	Pick one focus condition that frees the most capacity when improved	Pareto bar of total bed days by medical_condition

2	Inside the chosen condition, which admission_type has the longest median and p90 length_of_stay?	Bed Management	Target the admission pathway, creating the longest stays	Clustered bars of median and p90 LOS by admission_type
3	For that condition, which age group contributes the largest share of total bed days?	ICU Nurse Manager	Tune staffing and discharge planning to the cohorts that occupy beds the longest	Stacked column of total bed days by age group
4	On which admit_day_week do admissions and new bed days start the highest?	Operations	Staff around predictable weekly spikes to speed turnover	Column chart of admissions and total bed days by admit_day_week
5	If we reduce the p90 LOS by 15 percent for the target condition, how many bed days per week will we free?	COO / Operations	Turn process changes into a clear capacity impact for planning and ROI	KPI card for weekly bed days freed with a what-if slider

Methodology

Cleaning & Privacy:

- **Duplication & Consistency:** Resolved duplicate admissions by consolidating inconsistent values and removing redundant rows. Reset the dataset index and dropped non-essential columns to keep the model lean.
- **Standardized Columns:** Normalized categorical fields like gender, blood type and cleaned strings (removed white spaces, unified casing, converted empty strings to 'Unknown'). Enforced correct data types across keys and measures
- **Dates & LOS in SQL:** Analyzed admission/discharge dates and calculated Length of stay (LOS) in SQL for accuracy and reproducibility. Stored LOS as whole days.

- Identity Protection (HIPAA-aligned):
 - Created stable, non-reversible patient IDs by SHA-256 hashing patient names.
 - Issued unique doctor IDs and standardized their format
 - Any residual name fields were encrypted; only the minimum necessary fields were retained for analysis.

Data & Modelling

To make the analysis fast, explainable, and slicer friendly, I reshaped the source data into a classic star schema. The grain is one row per admission in a central fact table; everything descriptive (Who the patient is, what the condition label is, when the admission happened, which pathway, etc.) lives in small dimensions. This separation makes joins predictable, prevents duplication, and lets Power BI slicers behave deterministically because filters flow one way from dimension to fact.

A few design choices matter for this project's questions. Length of stay (LOS) is computed in SQL and stored on the fact, so percentile math (median/P90) happens at the right grain. Data logic is handled by a proper data dimension (with weekday and a numeric sort key), and I use tiny helper dims only when I need stable ordering for pre-aggregated CSVs. Finally, to keep slicers 'light' on the curated Q1-Q5 CSVs, I relate dimensions directly to those CSV tables (single direction) and do not connect the CSVs back to the fact to avoid ambiguous filter paths while keeping the page interactive.

Core Facts

- fact_admissions: admission_id, patient_id, doctor_id, medical_condition, insurance_id, admission_type, admission_date_sk, discharge_date_key, length_of_stay, billing_amount.

Dimensions

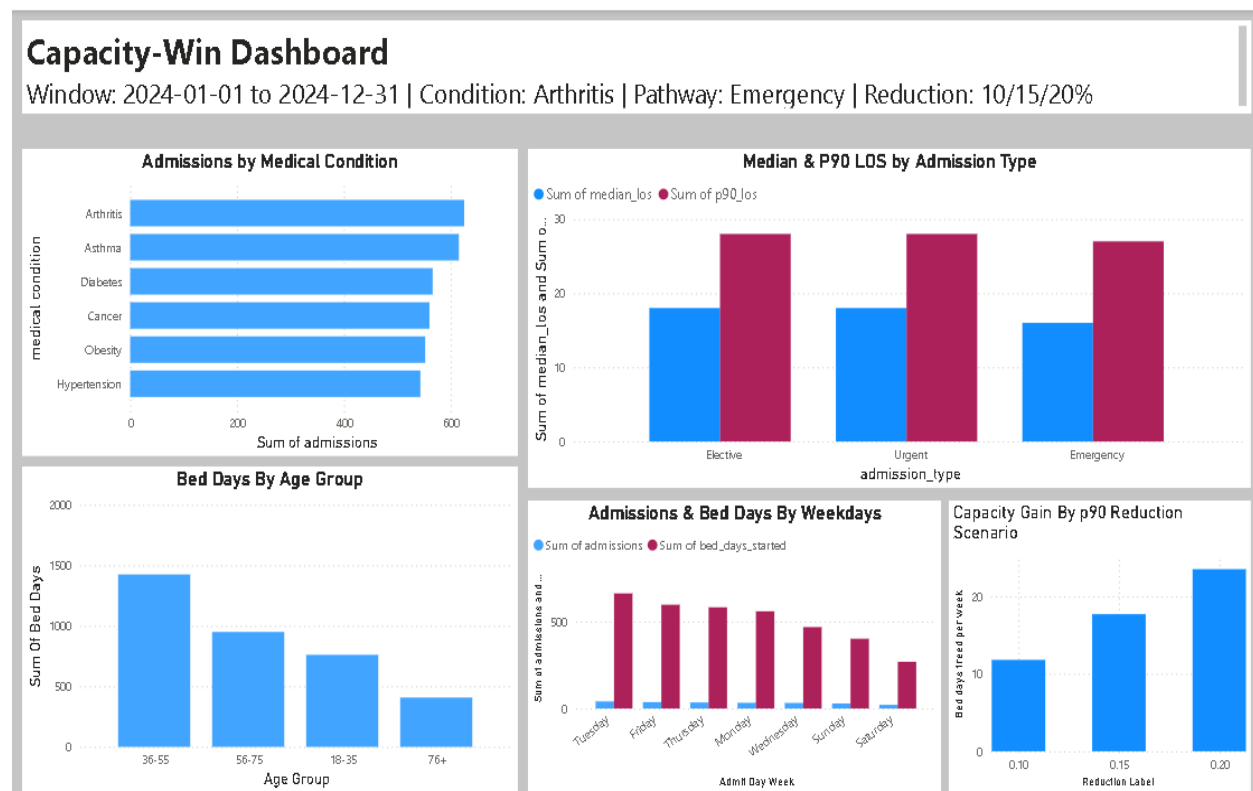
- Dim_date
- Dim_condition
- Dim_admission_type
- Dim_patient
- Dim_doctor
- Dim_insurance

By tying each question to the schema this way, the analysis stays both comprehensive (because all relevant dimensions are included) and consistent (because everything traces back to a single fact table).

Analytics Steps

- Q1. Condition Focus: sum total bed days and admissions by medical condition
- Q2. Long stay tail by pathway: Within the chosen condition, compute median (P50) and P90 LOS by admission_type.
- Q3. Who fills beds: Within the condition and chosen pathway, sum bed days by age_group and label share %.
- Q4. When pressure starts: Count admissions and sum bed days started by weekday of admission (sorted Mon-Sun).
- Q5. Capacity KPI:
 - Admissions/week = total admissions / weeks in window
 - Capacity win/week = admissions/week x p90_current x reductions%
 - Scenarios at 10%/15%/20%.

Power BI Dashboard



Results & Business Recommendations

Key Findings

- Focus Condition: Arthritis tops the list for total bed days, so it offers the biggest upside if we improve flow there first.
- Pathway choice (inside Arthritis): Emergency has a similar long stay tail to Elective and Urgent but higher admission volume, so small p90 cuts translate into the largest weekly capacity win.
- Who to plan around: Within Arthritis-Emergency, ages 36-55 drive the largest share of bed days (40% in our slice).
- When to staff: Tuesday is the peak for admissions and new bed days started (41 admissions; 663 bed days), so staff/diagnostics should be sequenced to handle that mid-week bulge.

Recommendation:

- Start a 2-week pilot in Arthritis – Emergency, designed around 36-55, and fortify Tuesday-Thursday throughput (tests, consults, step-downs) to shave the P90 tail and free capacity fastest.

What I would do with more time:

- Add severity/comorbidity and discharge destination to separate clinical vs system delays.
- Automate a what-if slider (10/15/20%) and weekly KPI trend in the report.
- Build a lightweight ETL so the dashboard refreshes with fresh data.

Code Snapshots

Question 1:

Which medical condition accounts for the most total bed days in the selected period?

```
set @start_date = '2024-01-01';
set @end_date = '2024-12-31';
select
    dc.medical_condition,
    SUM(fa.length_of_stay) as total_bed_days,
    count(*) as admissions
from hospital.fact_admissions as fa
join hospital.dim_condition as dc
    on dc.medical_condition = fa.medical_condition]
join hospital.dim_date as dd
    on dd.date_sk = fa.admit_date_sk
where dd.full_date between @start_date and @end_date
group by dc.medical_condition
order by total_bed_days desc;
#quick sanity check
select
    COUNT(*) as admissions,
    SUM(va.los_days) as total_los
from hospital.vw_admissions va
where va.admission_year = 2024;
```

Question 2:

Inside the chosen condition, which admission_type has the longest median and p90 length_of_stay?

```
set @start_date = "2024-01-01";
set @end_date = "2024-12-31";
set @condition = "Arthritis";
set @min_n = 20;
select
    t.admission_type,
    count(*) as admissions,
    min(case when t.cume >= 0.50 then t.los end) as median_los,
    min(case when t.cume >= 0.90 then t.los end) as p90_los
from(
    select
        COALESCE(NULLIF(TRIM(fa.admission_type), ''), 'unknown') as
        admission_type,
        fa.length_of_stay as los,
        CUME_DIST() OVER(
            partition by COALESCE(NULLIF(TRIM(fa.admission_type), ''),
            'unknown')
            order by fa.length_of_stay
        ) as cume
    from hospital.fact_admissions fa
    join hospital.dim_date dd
        on dd.date_sk = fa.admit_date_sk
    where dd.full_date between @start_date and @end_date
        and fa.medical_condition = @condition
) as t
group by t.admission_type
having count(*) >= @min_n
order by p90_los desc, admissions desc;
```

Question 3:

For that condition, which age group contributes the largest share of total bed days?

```
set @start_date = '2024-01-01';
set @end_date = '2024-12-31';
set @condition = 'Arthritis';
set @admission_type = 'Emergency';
select
    g.age_group,
    g.bed_days,
    g.admissions,
    round(100 * g.bed_days / SUM(g.bed_days) over (), 1) as share_pct
from (
    select
        coalesce(nullif(trim(dp.age_group), ''), 'unknown') as age_group,
        sum(fa.length_of_stay) as bed_days,
        count(*) as admissions
    from hospital.fact_admissions fa
    join hospital.dim_date dd
        on dd.date_sk = fa.admit_date_sk
    join hospital.dim_patient dp
        on dp.patient_id = fa.patient_id
    where dd.full_date between @start_date and @end_date
        and fa.medical_condition = @condition
        and fa.admission_type = @admission_type
    group by coalesce(nullif(trim(dp.age_group), ''), 'unknown')
) g
order by share_pct desc, g.bed_days desc;
```

Question 4:

On which admit_day_week do admissions and new bed days start the highest?

```
set @start_date = '2024-01-01';
set @end_date = '2024-12-31';
set @condition = 'Arthritis';
set @admission_type = 'Emergency';
select
    g.admit_day_week,
    g.dow_num,
    g.admissions,
    g.bed_days_started,
    round(100 * g.admissions / sum(g.admissions) over (), 1) as admissions_share_pct,
    round(100 * g.bed_days_started / sum(g.bed_days_started) over (), 1) as
    bed_days_share_pct
from (
    select
        dayname(dd.full_date) as admit_day_week,
        weekday(dd.full_date) as dow_num,
        count(*) as admissions,
        sum(fa.length_of_stay) as bed_days_started
    from hospital.fact_admissions fa
    join hospital.dim_date dd
        on dd.date_sk = fa.admit_date_sk
    where dd.full_date between @start_date and @end_date
        and fa.medical_condition = @condition
        and fa.admission_type = @admission_type
    group by admit_day_week, dow_num
) g
order by g.bed_days_started desc, g.admissions desc;
```

Question 5:

If we reduce the p90 LOS by 15 percent for the target condition, how many bed days per week will we free?

```
set @condition = 'Arthritis';
set @admission_type = 'Emergency';
set @reduction_pct = 0.15;
select
    s.admission_type,
    s.admissions,
    round(s.admissions / ((datediff(@end_date, @start_date) + 1) / 7.0), 2) as
    admissions_per_week,
    s.p90_los as p90_current,
    round(s.p90_los * (1 - @reduction_pct), 0) as p90_target_days,
    round(
        (s.admissions / ((datediff(@end_date, @start_date) + 1) / 7.0)) *
        (s.p90_los * @reduction_pct),
        1
    ) as bed_days_freed_per_week
from (
    select
        coalesce(nullif(trim(t.admission_type), ''), 'unknown') as admission_type,
        count(*) as admissions,
        min(case when cume >= 0.90 then t.length_of_stay end) as p90_los
    from (
        select
            fa.admission_type,
            fa.length_of_stay,
            cume_dist() over (
                partition by coalesce(nullif(trim(fa.admission_type), ''),
                'unknown')
                order by fa.length_of_stay
            ) as cume
        from hospital.fact_admissions fa
        join hospital.dim_date dd
            on dd.date_sk = fa.admit_date_sk
        where dd.full_date between @start_date and @end_date
            and fa.medical_condition = @condition
            and fa.admission_type = @admission_type
    ) t
    group by coalesce(nullif(trim(admission_type), ''), 'unknown')
) s;
```

Question 5a:

If we cut p90 by 10/15/20%, how many bed-days/week do we free? → **KPI** for capacity and ROI.

```
SET @start_date = '2024-01-01';
SET @end_date = '2024-12-31';
SET @condition = 'Arthritis';
SET @admission_type = 'Emergency';
SELECT
  s.admission_type,
  s.admissions,
  ROUND(s.admissions / ((DATEDIFF(@end_date,@start_date)+1)/7.0), 2) AS
admissions_per_week,
  s.p90_los AS p90_current,
  sc.reduction_pct,
  CONCAT(ROUND(sc.reduction_pct*100, '%') AS reduction_label,
  ROUND(s.p90_los * (1 - sc.reduction_pct),0) AS p90_target_days,
  ROUND(
    (s.admissions / ((DATEDIFF(@end_date,@start_date)+1)/7.0)) * (s.p90_los *
    sc.reduction_pct),
    1
  ) AS bed_days_freed_per_week
FROM (
  SELECT
    COALESCE(NULLIF(TRIM(t.admission_type), ''), 'unknown') AS admission_type,
    COUNT(*) AS admissions,
    MIN(CASE WHEN cume >= 0.90 THEN t.length_of_stay END) AS p90_los
  FROM (
    SELECT
      fa.admission_type,
      fa.length_of_stay,
      CUME_DIST() OVER (
        PARTITION BY COALESCE(NULLIF(TRIM(fa.admission_type), ''), 'unknown')
        ORDER BY fa.length_of_stay
      ) AS cume
    FROM hospital.fact_admissions fa
  JOIN hospital.dim_date dd
    ON dd.date_sk = fa.admit_date_sk
  WHERE dd.full_date BETWEEN @start_date AND @end_date
    AND fa.medical_condition = @condition
    AND fa.admission_type = @admission_type
  ) t
  GROUP BY COALESCE(NULLIF(TRIM(admission_type), ''), 'unknown')
) AS s
CROSS JOIN (
  SELECT 0.10 AS reduction_pct
  UNION ALL SELECT 0.15
  UNION ALL SELECT 0.20
) AS sc
ORDER BY sc.reduction_pct;
```